1.

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

1. Devices That Use SMPS and UPS

SMPS (Switched Mode Power Supply) and **UPS (Uninterruptible Power Supply)** are used to provide power to electronic devices. SMPS converts electrical power into the required voltage, while UPS provides temporary backup power during a power failure.

A. Three Devices That Use SMPS

S.No.	Device	Why It Needs SMPS
1	Desktop Computer	SMPS converts AC power into the required DC voltages to operate the motherboard, CPU, hard drive, and other components.
2	Laptop Charger	SMPS converts AC power from the wall socket into suitable DC power to charge the laptop battery and operate the laptop.
3	LED Monitor	SMPS converts the input power into the appropriate voltage required to operate the monitor's electronic components and display.

B. Two Devices That Use UPS

S.No.	Device	Why It Needs UPS
1	Desktop Computer	A UPS provides temporary backup power during a power cut, allowing users to save their work and shut down the computer safely.
2	Wi-Fi Router	A UPS keeps the router running during a power failure, helping maintain internet connectivity as long as the network connection remains available.

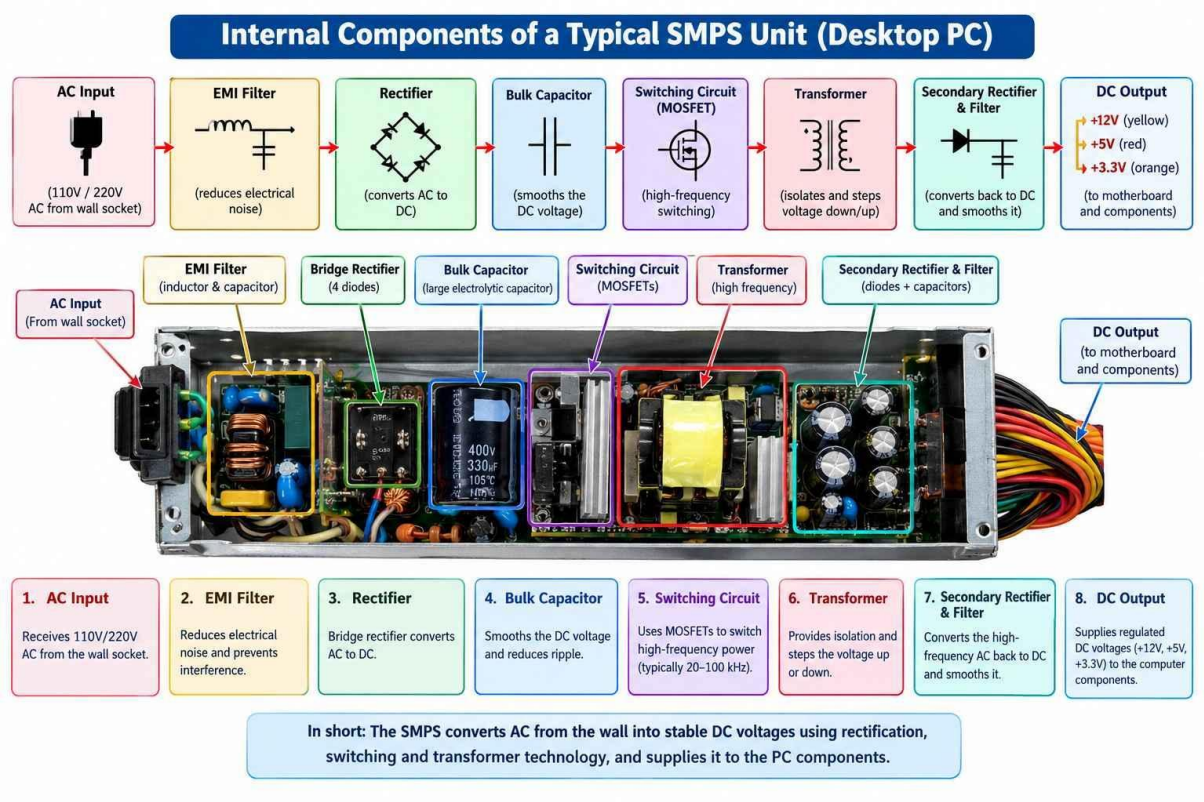
2.

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

S.No.	Component	Function
1	AC Input	Receives AC electrical power from the wall socket.
2	EMI Filter	Reduces electrical noise and electromagnetic interference.
3	Rectifier	Converts AC power into DC power.
4	Bulk Capacitor	Smooths the rectified DC voltage.
5	Switching Circuit (MOSFET)	Switches the electrical power at high frequency for efficient conversion.
6	Transformer	Provides electrical isolation and adjusts the voltage to the required level.
7	Secondary Rectifier and Filter	Converts the transformer's high-frequency output into DC and smooths it.

S.No.	Component	Function
8	DC Output	Supplies regulated DC voltages, such as +12V, +5V, and +3.3V, to computer components.



3.

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



Types of Power Connectors, Uses, and Safety Tips

S.No.	Connector Name	Devices It Powers	One Safety Tip
1	**ATX 24-pin Motherboard Connector**	Supplies the main power to the motherboard and its components.	Switch off and unplug the computer before connecting or disconnecting it. Press the locking clip to remove the connector safely.
2	**SATA Power Connector**	Supplies power to SATA SSDs, HDDs, and optical drives.	Align the L-shaped connector correctly with the port. Never force it into the socket.
3	**Molex 4-pin Connector**	Powers older hard drives, case fans, and other compatible accessories.	Disconnect the power supply and pull the connector by its body, not by the wires.

S.No.	Connector Name	Devices It Powers	One Safety Tip
4	**PCIe 6/8-pin Connector**	Supplies additional power to graphics cards that require it.	Use the correct PCIe power cable, not the CPU power cable, and ensure the connector is securely fitted.

Brief Explanation

1. ATX 24-pin Connector:
This is the main power connector that connects the SMPS to the motherboard. It provides the required electrical power for motherboard components.

2. SATA Power Connector:
This is a flat, 15-pin connector used to provide power to SATA storage devices, such as SSDs and hard disk drives.

3. Molex 4-pin Connector:
This connector has four large pins and is commonly used with older storage devices, fans, and other compatible accessories.

4. PCIe 6/8-pin Connector:
This connector provides additional power to a dedicated graphics card. Some graphics cards require a 6-pin or 8-pin connector, while others may not require a separate power cable.

4.

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

SMPS (Switched Mode Power Supply) supplies electrical power to the motherboard, CPU, graphics card, and storage devices in a gaming PC. Different components require specific power cables and connectors.

Step-by-Step Procedure

Step 1: Prepare the SMPS

- Switch off and unplug the power supply from the wall socket.
- Install the SMPS in the computer case and secure it with screws.
- Identify the required power cables.

Step 2: Connect the Motherboard

- **Cable:** 24-pin ATX power cable.
- Connect the 24-pin connector from the SMPS to the motherboard's main power socket.
- Ensure that the locking clip clicks into place.

Purpose: Supplies the main power to the motherboard and its components.

Step 3: Connect the CPU Power Connector

- **Cable:** 4+4-pin CPU/EPS power cable.
- Connect it to the CPU power socket, usually located near the top of the motherboard.
- Ensure the connector is securely fitted.

Purpose: Supplies power to the CPU through the motherboard's voltage-regulation circuitry.

Step 4: Connect the Graphics Card

- **Cable:** PCIe 6-pin or 8-pin power cable, or the appropriate 12V-2x6 cable for compatible graphics cards.
- Connect the correct power cable to the graphics card if it requires additional power.
- Check that the connector is fully inserted.

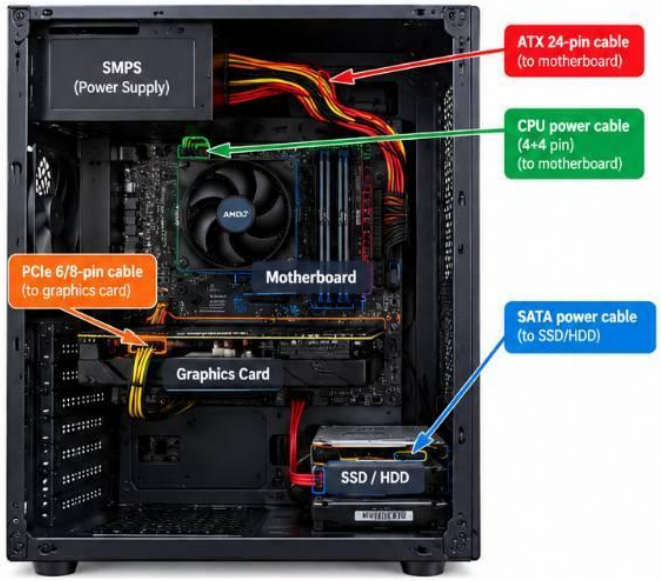
Purpose: Supplies additional power required by a dedicated gaming graphics card.

Step 5: Connect the Storage Devices

- **Cable:** 15-pin SATA power cable.
- Connect one end to the SMPS's SATA power connector and the other end to the SATA SSD or HDD.
- For a SATA drive, also connect a SATA data cable between the drive and the motherboard.

Connecting SMPS to Motherboard, Graphics Card and Storage Devices in a Gaming PC

The SMPS (Power Supply) provides power to all the main components of your gaming PC through different cables and connectors.



1 Connect the ATX 24-pin to the Motherboard

- Use the 24-pin ATX power cable from the SMPS.
- Plug it into the 24-pin connector on the right side of the motherboard.
- Make sure the latch clicks and the connector is fully seated.

2 Connect the CPU Power (4+4 pin)

- Use the 4+4 pin CPU power cable (often labeled "CPU" or "EPS").
- Plug it into the 8-pin (or 4+4 pin) connector on the top-left of the motherboard.
- This powers the CPU and helps with stable performance.

3 Connect the PCIe Power to the Graphics Card

- Use the PCIe 6/8-pin cable (labeled "PCIe" or "VGA").
- Plug it into the power connector on the graphics card.
- Most high-performance cards need 6-pin or 8-pin connectors.

4 Connect SATA Power to Storage Devices

- Use the SATA power cable (flat, 15-pin connector).
- Connect it to the SSD or HDD.
- The data cable (SATA) also connects the drive to the motherboard.

5 Check and Organize

- Ensure all connectors are firmly connected.
- Route and tie the cables for better airflow and neatness.
- Double-check that the system is powered off before making changes.

Cables and Connectors Used

ATX 24-pin (Motherboard)



Connects to: Motherboard
Cable type: 24-pin ATX cable
Common label: ATX / MB

CPU 4+4 Pin



Connects to: Motherboard
Cable type: 4+4 pin EPS cable
Common label: CPU

PCIe 6/8 Pin



Connects to: Graphics Card
Cable type: 6 or 8 pin PCIe cable
Common label: PCIe / VGA

SATA Power



Connects to: SSD / HDD
Cable type: SATA power cable
Common label: SATA

Quick Summary

Component	Cable/Connector Used	Purpose
Motherboard	ATX 24-pin	Provides main power to motherboard and its components.
CPU	4+4 pin (EPS)	Powers the processor (CPU).
Graphics Card	6/8-pin PCIe	Supplies additional power to the GPU.
Storage (SSD/HDD)	SATA power (15-pin)	Powers SATA drives.

Safety Tips:

- Always turn off and unplug the PC before connecting or disconnecting power cables.
- Do not force connectors — check the shape and alignment.
- Use the correct cable and connector for each component.
- Keep cables organized for better airflow and cooling.